



Monetary policy and risk taking



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ABSTRACT

We assess the effects of monetary policy on bank risk to verify the existence of a risk-taking channel – monetary expansions inducing banks to assume more risk. We first present VAR evidence confirming that this channel exists and is particularly significant on the bank funding side. Then, to rationalize this evidence we build a macroeconomic model where banks subject to runs endogenously choose their funding structure (deposits vs. capital) and risk level. A monetary expansion increases bank leverage and risk. In turn, higher bank risk in steady state increases asset price volatility and reduces equilibrium output.

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1. Introduction

It is widely believed that the 2007 financial crisis originated from mis-incentives in the financial markets leading to excessive leverage and risk-taking by financial institutions. High liquidity and persistently low interest rates, combined with lenient bank supervision, allegedly induced banks to finance an increasing volume of risky assets – largely in the real estate sector – by means of cheap short-term funding. This line of argument calls into question the links between monetary policy and financial risk-taking. Largely neglected prior to the crisis – with some notable exceptions, mentioned below – such links are now increasingly discussed,² but two elements are missing to provide a foundation to the argument: realistic macroeconomic models that endogenize risk taking behavior and relate it to monetary policy, and time-series evidence documenting this relation.

We move in that direction in two ways. First, we look at time series evidence on the link between monetary policy and risk taking. The empirical literature has been confined to survey and panel data evidence; no aggregate-level time series tests are available. Tests involving aggregate dynamics are important because interest rate changes are likely to influence the banks' balance sheet risk in different ways at different time lags: in the short run, risk is likely to be positively correlated with interest rates, but in the medium to long run this relation may be inverted if the risk-taking channel dominates. Second, we propose a model, based on [Diamond and Rajan \(2000, 2001\)](#), that rationalizes such channel. In our model bank

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² For a recent review of the debate see [Dell'Ariccia et al. \(2011\)](#).

managers endogenously choose between two sources of funding, uninsured short-term liabilities (we call them, in short, deposits) and bank capital, to finance risky investment projects. Bank managers have an informational advantage on the projects and act as relationship lenders on behalf of the two outside financiers of the bank, namely depositors and capitalists. Fundamental bank runs arise as a discipline device: when a run materializes, banks must liquidate projects. This both affects the payoff structure among the three bank stakeholders (depositors, capitalists, managers) and entails an aggregate resource loss. Low policy rates reduce the cost of short term finance to banks and, if protracted, provide an implicit guarantee that indirectly impairs market discipline. When rates are low, banks substitute bank capital with deposits, raising bank riskiness (probability of a bank run). Since the probability of bank runs is endogenous, the model can account for the evolution of bank risk in relation to monetary policy and the business cycle.

A noteworthy feature of our model consists in embedding fundamental bank runs into a macroeconomic model for policy analysis. [Diamond and Dybvig \(1983\)](#) modelled panic based banks runs in a partial equilibrium and static context: they analyzed panic runs triggered by liquidity shocks on depositors. Since then, the banking literature has evolved. On the one hand, empirical evidence³ has documented a correlation between banks' runs and changes in fundamentals. On the other, the notion of purely panic-based run does not lend itself easily to policy analysis, because of the difficulty of pinning down an endogenous probability of bank runs (there are two rational equilibria, each with equal probability). For this reason the theoretical banking literature moved towards considering fundamental and information-based bank runs, ultimately triggered by bad news on investment returns. We follow this latter notion of bank run, embedding it into a macroeconomic model and analyzing its interaction with monetary policy.⁴

We obtain three sets of novel results. First, our time series evidence, based on a standard VAR augmented with various sources of bank risk, supports the notion that monetary policy influences risk in the banking sector by changing the bank's funding structure, as well as the riskiness of its assets. These results are robust to different proxies of bank risk and different VAR identification strategies. Second, we propose a model that rationalizes the risk taking channel on the funding side. Our model shows the mechanisms through which an expansionary monetary policy raises bank leverage and risk (and a contractionary policy does the opposite), by inducing banks to substitute short term risky funding instruments for capital. The effects of the monetary expansion on output and inflation are the conventional ones – they both rise – but they are milder than in a corresponding model without banks; a dampening of monetary policy transmission occurs because risk-taking by banks is contractionary, hence it compensates in part the expansionary first-round effect. Similar effects occur under a positive productivity shock, due to the fact that monetary policy becomes more expansionary, as it targets expected inflation. Third, we also discuss the effects of projects riskiness on the long run levels and the volatility of output and assets prices. The literature found extensive evidence that an increase in such riskiness raises the volatility of output and reduces its long run level⁵ as well as raises the volatility of asset prices and reduces their long run level.⁶ Our model confirms those links, but highlights a new channel that stems from the endogenous formation of risk: when investment project risk increases, and as investors become aware of such increase, more bank runs occur. This raises the volatility of bank funding and investment and lowers production prospects in the long run due to the resource costs of projects' liquidation.

The paper is organized as follows. In [Section 2](#) we briefly review some recent literature on the risk taking channel of monetary policy. In [Section 3](#) we present time-series evidence on the transmission of monetary policy on bank risk in the US. In [Section 4](#) we review some macro-finance literature and relate our theoretical model to that by highlighting novelties and differences. In [Section 5](#) we present our macroeconomic model with bank runs. In [Section 6](#) we analyze the model and its quantitative properties, mostly in relation to our time series evidence. Finally, [Section 7](#) concludes. Appendices and tables follow.

2. Recent empirical evidence

The surge of interest for the implications of monetary policy on financial risks after the recent crisis contrasts sharply with the virtual absence of any reference to risk⁷ in the earlier literature on monetary policy transmission. The classic 1995 survey by Mishkin, Taylor and others in the *Journal of Economic Perspectives* ([Mishkin, 1995](#)) hardly mentions bank and financial risks at all. In the multi-country empirical study of monetary transmission in the euro area conducted by the Eurosystem central banks, dated 2003,⁸ indicators of bank risk are actually used in the econometric estimates of the “lending channel,” but only to measure how changes in certain structural characteristics of the banking sector affect the strength of the transmission, not because monetary policy may itself influence those characteristics.⁹

In a different context, however, other authors had stressed the potential importance of the link between monetary policy and financial risks well before the onset of the financial crisis. [Allen and Gale \(2000\)](#) had provided a theoretical underpinning for these ideas by showing how leveraged positions in asset markets create moral hazard. In their paper leveraged investors

³ See among others [Kaminsky and Reinhart \(1999\)](#) and [Calomiris and Mason \(2003\)](#) for links between bank runs and fundamentals.

⁴ See [Diamond and Rajan \(2000, 2001\)](#). In a companion paper, [Angeloni and Faia \(2013\)](#) provide normative analysis within a similar model, also showing that the model matches the main macroeconomic and banking business cycle features.

⁵ See [Bloom \(2009\)](#).

⁶ See for instance [Bae et al. \(2007\)](#).

⁷ As explained earlier by risk here we mean mainly indicators of endogenous formation of risk, not merely exogenous financial shocks.

⁸ See [Angeloni et al. \(2003\)](#).

⁹ Recently a Minskian view of endogenous risk formation has been proposed in [Assenza and Delli Gatti \(2013\)](#) but in a model with firms' default.

can back-stop losses by defaulting, and this makes asset prices deviate from fundamentals. The link with monetary policy, clarified in later work by [Allen and Gale \(2007\)](#), consists in the fact that aggregate credit developments in the economy are, at least partly, under the control of monetary authorities. [Borio and Lowe \(2002\)](#) described how asset market bubbles, leading to financial risk and instability, can develop in a benign macroeconomic environment, including high growth, low inflation, low interest rates and accommodative monetary policy.¹⁰

To help the subsequent analysis, it is useful to distinguish between two different channels through which risk-taking behavior can operate. The first refers to accumulation of excessive risk on the funding side. An expansionary monetary policy may affect the composition of bank liabilities, altering the mix of capital (plus other stable funding sources) and short term funding in favor of the latter. This channel operates in particular when short term rates are low and the yield curve upward sloping. The second channel is via changes in the degree of riskiness of the intermediary's asset side. In the presence of low and persistent interest rates levels, asset managers of banks and other investment pools may have an incentive to shift the composition of their investments towards a riskier mix (see for instance [Rajan, 2005](#)). Risk taking on the funding side may in fact initiate and amplify risk taking on the asset side: as banks can transfer risk to outside financiers, through higher leverage, their incentives toward riskier investments increase. Statistical and anecdotal information confirm that financial institutions of various sorts (banks, conduits and SIVs, investment funds, insurance companies, etc.) on both sides of the Atlantic became riskier, in the pre-crisis years, due to excessive leverage.

The empirical evidence on these transmission channels has grown fast in recent times. So far the analysis has focused on micro-survey data and on a panel dimension. [Maddaloni and Peydró Alcalde \(2010\)](#) use evidence from a euro area lending survey to see whether monetary policy influences the lending practices of banks. The survey allows them to distinguish between supply related factors (i.e. linked to bank-specific conditions) and demand related ones (i.e. depending upon borrowers' conditions). The authors use a panel regression to link the survey results to alternative indicators of monetary policy. The proxy for monetary policy has consistently significant effects: a monetary expansion leads to lower credit standards, for corporate as well as personal loans. Moreover, the longer a given policy stance lasts, the more effect it seems to have.

Another paper ([Altunbas et al., 2009](#)) uses a more comprehensive sample and a different measure of bank risk. They consider over 600 listed European banks, in 16 countries for which Moody's KMV has computed expected default frequencies (EDF hereafter). EDFs, expressing market perceptions of the default probability at a given time horizon, are a widely used measure of bank risk, shown to have predictive power in many cases. EDFs are obtained by translating, with a model, several market and balance sheet indicators into a single measure, a time-varying probability of default at a specific time horizon. The authors make this the dependent variable in a panel regression, that includes a variety of explanatory factors – macroeconomic variables, market data, other bank characteristics – as well as monetary policy. The results suggest that a decrease of short term rates reduces overall bank risk in the short run – as one would expect, since lower interest rates on impact improve the financial condition of borrowers via changes in the value of collateral – but increases it over time. A plausible interpretation is that while the risk of *existing* loans is *positively* related to the level of the policy-determined interest rate, the risk of loans that are *issued subsequently* to the increase of such rate is *negatively* related to it, because the lending behavior of the bank changes. Measures of the average risk of loans combine the two elements, hence one tends to observe a switch in sign between the short and the long run.

In view of the possibility of these interacting dynamic effects, empirical evidence of the risk taking channel on macro-time series can be of considerable interest, but has so far been missing. In the next session we move a step forward in the direction of testing the risk taking channel at macro-level.¹¹

3. Time series evidence

In this section we report new time series results on the effect of monetary policy on bank risk. We regard this evidence as suggestive, and use it mainly to provide empirical backing to the model-based analysis that follows. In this section we refer to both channels through which monetary policy can affect bank risk, namely, via the banks' funding and lending behavior. In the rest of the paper we will focus on the first channel that our model is best suited to analyze.

We use a standard orthogonalized VAR model, with quarterly US data over the period 1980 Q1 to 2008 Q4. We exclude the period after 2008, where our monetary policy indicator – the Federal Funds rate – is constant at zero, and the monetary policy stance is better described by other (non-standard) indicators. We adopt, with modifications, the specification by [Christiano et al. \(1999\)](#), also discussed in [Faust et al. \(2004\)](#). The original VAR includes, in the following order, the industrial production index, the consumer price index CPI, a commodity price index (to account for the global component of price dynamics), the federal funds rate, non-borrowed bank reserves and total bank reserves. The shock in the Federal Fund rate equation is considered the monetary policy shock. The identification assumption implicit in the ordering of the variables is that monetary policy reacts to prices and the real activity. We extend the original set up of [Christiano et al. \(1999\)](#) by adding

¹⁰ This seminal contribution was followed by a host of publications by economists at the Bank for International Settlements calling for the adoption of a “macroprudential approach” to financial stability including, notably, a response of monetary policy to asset prices.

¹¹ The aggregate time series perspective can be important also for two additional reasons. First, to verify how significant are these risk-inducing effects at the macro-level. Second, time series evidence allows us to consider the endogenous response of monetary policy: VAR evidence would indeed allow us to verify whether the endogenous response of monetary policy can neutralize or, on the contrary, encourage, risk taking behavior.

measures capturing bank funding risk, the riskiness of bank assets and an indicator of overall bank risk (all details on data definitions and charts of the variables we used are contained in [Appendix E](#)):

- **Funding risk:** To capture risks stemming from the funding structure of banks, we use measures of non-core bank liabilities ([Shin and Shin, 2010](#)). These liabilities, that have grown sharply in recent times, consist of short term revolving funding instruments like CDs, repos, asset backed instruments and the like, carrying a non-contingent contractual return and subject to roll-over risk. These funding sources are subject to sudden withdrawal if market confidence deteriorates. [Hahm et al. \(2013\)](#) show that, while retail deposits (core liabilities) are stable, non-core liabilities are more volatile, are associated with credit booms and have significant predictive power for currency and credit crises. According to [Shin and Shin \(2010\)](#), the difference between two measures of broad money, $M3 - M2$, captures well non-core liabilities and some aspects of wholesale bank funding. Therefore, we calculate our benchmark measure of funding risk by summing up the categories that make up the difference between $M3$ and $M2$, namely repos, large time deposits and assets of money market mutual funds.¹² We also conduct a number of robustness tests by looking at specific liabilities as, for example, repos.
- **Lending risk:** Capturing lending risk is not an easy task. Improvements in risk indicators might result from wrong beliefs and complacency by investors. For example, we now know that bank balance sheets were deteriorating in the years before the financial crisis, as banks were accumulating concentrated exposures to highly indebted households; yet market-based measures of bank default probability remained very low until 2007.¹³ While household leverage was increasing in the pre-crisis period, pointing to risks and vulnerabilities, it is hard to identify indicators as, for example, spreads or ratings, flagging risks in banks' assets stemming from the exposure to households. For these reasons, we decided to measure the lending risk by looking at the soundness of bank borrowers i.e. households and corporations. In our benchmark specification, we use the stock of debt of household and non-financial corporations.
- **Total bank risk:** This variable captures the risk components mentioned above, namely funding and lending risk. To measure bank risk we use the realised volatility of the Datastream US bank equity index. The realised volatility is calculated as the average daily absolute return of the index over each quarter.

The above measures of bank risk are meant to identify possible channel of transmission of monetary policy to bank risk, respectively via the liability side, the asset side and both sides of the balance sheet. In particular, we expect that, if there exist a “risk taking channel” of monetary policy running via the funding side, the first and last of the above proxies should decline when monetary policy is tightened. If instead a risk taking behavior exists only on the lending side, then the measures capturing non-financial sector leverage and overall bank risk should show a significant decline. If no risk taking channel to banks exists, none should be significant, if both channels exist, all effects should be significant.

As the inclusion of non-borrowed and total bank reserves turns out to be irrelevant for the impulse responses, the below results refer to a specification where bank reserves are dropped. Our final benchmark specification includes the following variables in this ordering: industrial production index, consumer price index, commodity price index, household debt (asset side risk), non-financial corporation debt (asset side risk), Federal Fund rate, difference between $M3$ and $M2$ (funding side risk), realized volatility of bank equities (overall bank risk). As in [Christiano et al. \(1999\)](#), the residual in the equation of the Fed Fund rate is considered as the monetary policy shock. The identification assumption implicit in the ordering of the variables is that monetary policy reacts to prices and the real activity and, possibly, to debt in the non-financial sector. Positioning the debt of the non-financial sector after the Federal Fund rate does not alter the results. We use an alternative identification strategy in the robustness analysis. All the variables included in the model are expressed in logs (with the exclusion of the Fed Fund rate and the total bank risk variable) and are described in the data appendix ([Appendix E](#)). The VAR is estimated with one lag, according to the Schwarz and the Hannan–Quinn information criteria.

[Fig. 1](#) shows the main results of the analysis in the form of dynamic impulse responses over a time window of four years (16 quarters). A contractionary monetary shock has the expected signs on real output (panel A): the industrial production index declines significantly reaching a trough after about two years. The reaction of the consumer price index (panel B) presents a small price puzzle on impact while it drops subsequently. The above results on the impact of monetary policy shocks on real variables and prices are in line with [Christiano et al. \(1999\)](#) and with other more recent studies that use alternative identification strategies for monetary policy shocks (for a review see [Barakchian and Cowe \(2010\)](#)).

Concerning measures of bank asset risk, the shock produces an immediate decline of both household debt (panel E) and non-financial corporation debt (panel D). The reduction peaks towards the end of the forecasting horizon. These results are consistent with the existence of a risk-taking channel on the asset side of banks. Regarding bank liabilities, the monetary contraction decreases the reliance of banks on volatile sources of funding (non-core liabilities). Funding risk starts to decline after about six quarters (panel C) while the reduction peaks around the end of the forecasting horizon. These findings are consistent with the above results for asset side risk and confirm the existence of a risk-taking channel on the funding side. Finally, note that our measure of overall bank risk (panel F) displays a response profile broadly consistent with that of our

¹² In the US, the publication of the monetary aggregate $M3$ has been discontinued in 2005, therefore computing the difference between $M3$ and $M2$ is not a feasible option. In order to overcome the unavailability of the simple difference between $M3$ and $M2$, we collect data on the components of the difference. The data on large time deposits, money market mutual fund assets and REPOs were collected from the US flow of funds dataset (L110, liabilities of depository institutions for REPOs and large time deposits; L206 for money market mutual fund assets).

¹³ This has been called the “paradox of financial instability” by [Borio and Drehmann \(2009\)](#).

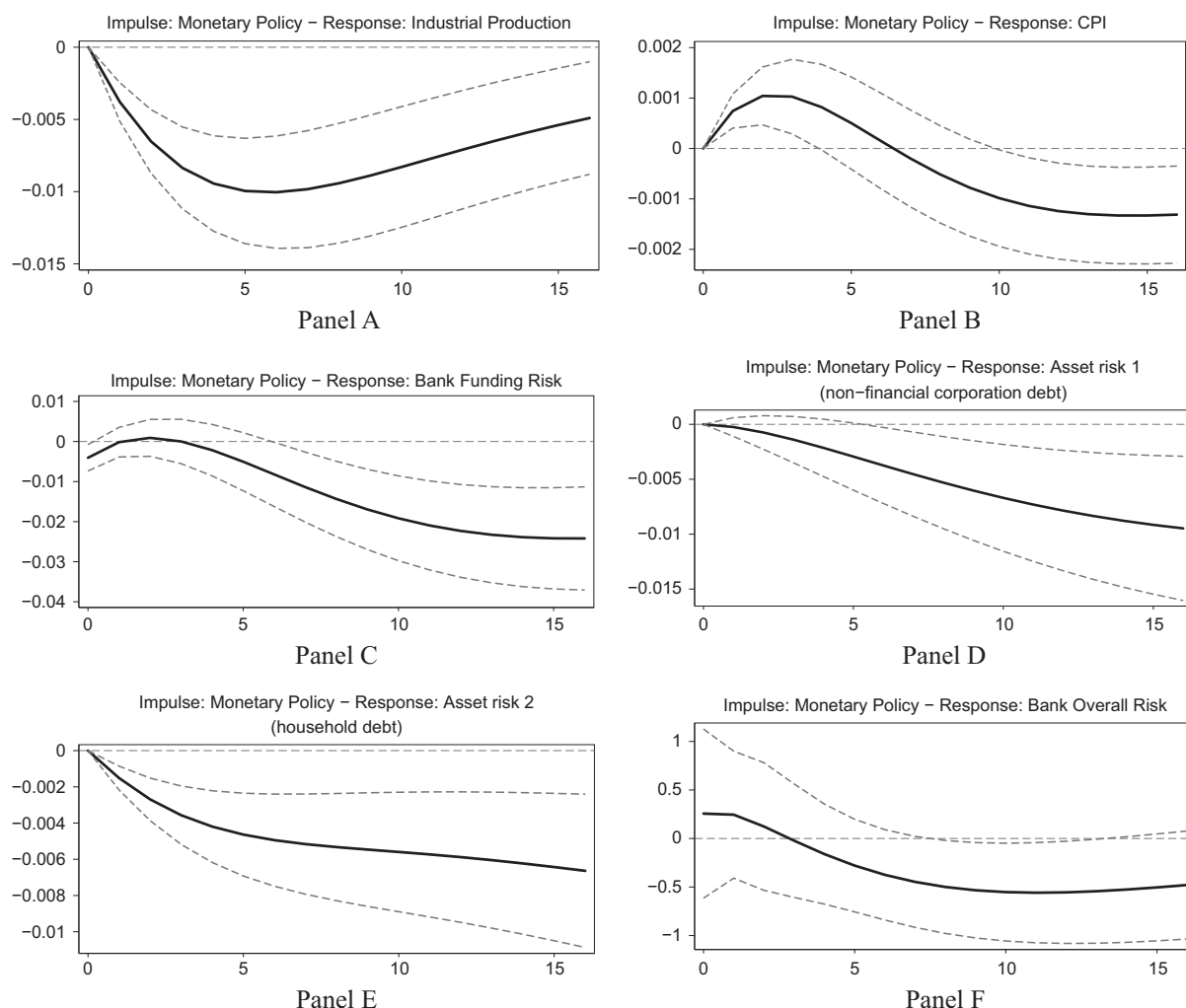


Fig. 1. Impulse response to monetary contraction on output (A), on prices (B), on bank funding risk (C), on bank exposures to firms (D), on bank exposures to households (E) and on bank overall risk (F). Dotted lines indicate 90% confidence intervals.

asset and funding risk measures; negative and significant at longer lags when the decrease in asset and funding risk is also large and significant.

We conducted a number of checks to verify the robustness of our results.

First, we use alternative variables for the business cycle (GDP, industrial production excluding construction, ISM index, the components of the ISM index measuring current economic conditions) and for prices (consumer price index excluding energy and food prices). In addition, we include the realized volatility of the S&P500 index to control for overall uncertainty and risk, which might affect our results regarding the bank overall risk. Our main results are not affected by these changes.

Second, we test whether our results are stable by changing the way we measure funding and asset side bank risk. For asset side risk we replace the stock of debt with leverage ratios. Specifically, we use the debt over GDP ratio for non-financial corporations, while we use both the debt to GDP and the debt to income ratios for households. For the funding risk, we use the stock of repos. Our main findings are confirmed when using these variables, although the statistical significance of the results for asset side risk is weaker for households.

Third, the results on funding risk and bank overall risks are confirmed in a setting where the VAR model is estimated using monthly data and using alternative variables to capture funding risk. In particular, the results hold when funding risk is measured either by looking at interbank liabilities or at all liabilities excluding retail deposits. As a measure of overall bank risk we alternatively used the expected default frequencies produced by Moody's KMV. Also in this setting the results remained stable.

A further robustness check was conducted on the identification strategy of the monetary policy shocks. As mentioned above, our main results are stable when using other plausible ordering of the variables in the VAR. In addition, we used an alternative measure of monetary policy shocks, namely the one calculated by [Romer and Romer \(2004\)](#). This measure is designed to overcome the bias that emerges when the intended change in the Fed Funds rate is measured with uncertainty

and when FOMC responds to beliefs about the future path of the economy which is unobserved in our VAR. We borrow the updated Romer and Romer shocks from [Barakchian and Cowe \(2010\)](#) and we estimate the VAR by placing the new monetary policy shock variable either first (assuming that this is an exogenous shock) or after output and price variables. All else remains as in our benchmark model. Our results are confirmed also in this setting.

4. Relation to the macro-finance literature

We have already discussed the links between our paper and the literature on the risk taking channel, which was so far located primarily in the areas of finance and micro-econometrics. Our paper has also some relation to the papers in the macroeconomic literature studying the relation between financial frictions and monetary policy. This literature examines how deviations from the Modigliani–Miller theorem result in the value and risk of the bank being affected by its funding structure. This has also implications for the transmission mechanism of policy and other shocks. Here we will briefly review the most important differences between our model and some papers in this literature.

First, our model introduces optimizing banks and focuses on the fragility of banks funding structure, rather than on the firms' lending relations with banks. On the contrary, papers in the traditional financial accelerator literature (see [Bernanke et al., 1999](#)) focus on the firms' lending frictions, while treating the intermediary as a pool of households' resources. In terms of transmission mechanisms there are not only similarities, but also important differences. Our model shares with the financial accelerator literature the presence of a “balance sheet channel:” the fall in the policy rates, by boosting asset prices, increases the value of balance sheets (banks' balance sheets in our model, firms' balance sheets in the traditional financial accelerator literature). Due to this channel, in the financial accelerator literature monetary policy expansions increase the value of firms, raise the value of collateral and reduce firms' default probability, hence boosting credit supply and aggregate demand (see [Bernanke et al., 1999](#)). As a result in the financial accelerator literature the firms' premium for external funding is typically counter-cyclical: as firms' default probability falls the premium for external funding falls as well.

Recent papers (see [Gertler and Karadi, 2011](#) or [Gertler and Kiyotaki, 2010](#)) have imported the financial accelerator mechanism on the banks' funding side. The effects of monetary policy expansions are similar. A reduction in the policy rate, by boosting assets prices, increases the value of banks' balance sheets and banks' capital, reduces overall banks' risk and reduces the spreads on banks' external funding. In those models banks' capital is pro-cyclical with respect to output and countercyclical with respect to risk, while banks' risk and spreads are counter-cyclical with respect to output.

In our model a reduction in the policy rate features an additional “risk taking channel”. Falls in the policy rates induce banks to increase the share of short term liabilities (at the expenses of banks' capital), therefore increasing the risk of bank runs and defaults as well as the related risk premium. In our model the risk premium (or the banks' external finance premium) is therefore pro-cyclical with respect to output and under monetary policy shocks. Moreover, in our model bank's capital is counter-cyclical with respect to output and pro-cyclical with respect to risk under standard productivity and monetary policy shocks. Bank's capital pro-cyclical with respect to risk is well grounded in empirical analyses (see among many others [Ayuso et al., 2004](#); [Estrella, 2004](#); [Jokipii and Milne, 2008](#)). The possibility of replicating this well known fact is one of the features of our model.

Another implication of the above transmission mechanism is that in our model a build-up of bank risk tends to be associated with a dampening of the business cycle. This effect is in line with the “volatility paradox,” according to which low volatility of output can be associated with a built up of risk (see [Brunnermeier and Sannikov, 2014](#) for a theoretical model generating this effect). In our model there is a two way causality. In times of stable and sustained growth the risk taking behavior of banks who leverage up induces an endogenous increase in aggregate risk; as aggregate risk brings about resource costs, the latter tend to dampen output in the long run. This mechanism is absent in the literature with credit frictions on firms' lending, which instead generally features amplification of macroeconomic shocks in face of negative demand or policy shocks, firms tighten investment strongly due to both the reduced returns and the increase in the lending spreads.

Finally, other papers in the macro-finance literature introduce a lending spread between the interest rate applicable to household savings and the interest rate at which households borrow. One example is [Curdia and Woodford \(2009\)](#) and the companion paper [Curdia and Woodford \(2009\)](#). In their model banks optimize the stream of dividends to households. Risk is introduced in their model through an exogenous shock (either a financial intermediation productivity shock or an exogenous premium introduced to compensate for the possibility of firms' default), which can also induce default in some loans. Our model in contrast focuses on endogenous determination of banks' default risk.

5. A macroeconomic model with fundamental bank runs

The financial side of our model features banks with an endogenous funding choice and endogenous risk of bank runs. Banks receive two sources of funding: demand deposits and bank capital. These two funding sources are combined to finance risky projects. Demand deposits are subject to a non-contingent service constraint, which exposes banks to runs. If no run occurs, bank capitalists receive a rent, which compensates them for the risk of losses in the run states. The bank is administered by a bank manager, a “relationship lender” who by lending acquires a superior knowledge on the project's quality. The manager chooses the optimal funding structure (the optimal shares of demand deposits and bank capital) to maximize total expected returns to outside financiers. The bank manager's superior skills effectively create a moral hazard problem since the manager is tempted to withhold its technology, forcing a costly liquidation of the loan. Those incentives are disciplined in two ways. First, depositors can threat a

run, a feature that effectively works as a discipline device. Second, the contractual agreement between the bank manager and the outside financiers takes the form of a bargaining arrangement in which the bank manager receives a fraction (depending on the relative bargaining power) of the total returns: ex post bank managers have thus incentives to maximize total expected returns from the project.

Notice that in our model deposits are not traditional retail deposits, which usually are largely insured. They are instead uninsured short term funding instruments (for example, asset-backed securities or repos), yielding a contractual non-contingent return set ex ante, and subject to run in the form of roll-over risk. These instruments are ultimately, directly or indirectly (through saving pools), owned by households. In the US, the outstanding volume of these funding instruments has grown enormously in the years preceding the crisis, though precise data on its exact size do not exist (Gorton and Andrew, 2012). Considering repos alone, a large subset of the total, estimates by Hördahl and King (2008) suggest that their volume at the beginning of 2008 amounted to over 10 trillion dollars, of which over 3 trillion directly in the balance sheet of the banks. Other short-term funding instruments, such as ABSs and MBSs, may add roughly another equivalent amount in total.¹⁴ This compares with total liabilities of commercial banks of around 10 trillion, of which 6 in the form of deposits.¹⁵ The right estimate of the potential impact of the repo market on banks is probably somewhere in between, because non-bank repos (mainly made by primary dealers) can also affect banks indirectly. Whatever the exact measure, it is clear that a run on the repo market can have devastating effects on banks and the financial sector, as the 2007–2008 experience shows.

In our model bank risk affects households/investors returns in two ways. First, the expected return to depositors decreases, to an extent determined endogenously by the probability of run and by the expected loss per unit of deposits conditional on a run, g_t . Second, there is a resource costs for the whole economy due to the fact that, when a run occurs, banks have to liquidate the projects in advance, hence part of the projects' proceeds is lost.

Notice also that our model only focuses on risk taking on the banks' funding side: bank risk in our model is given by the endogenous probability of banks' runs on short term liability. We deliberately focused on this for three reasons. First, given the complexity of the transmission mechanisms in macroeconomic models with banking, focusing on one dimension of the risk taking channels allows us to provide a more clear cut discussion. Second, our time series evidence above suggested that the risk taking channel on the funding side plays a distinct and more significant role, relative to the risk taking channel on the asset side. Third, much of the recent literature has emphasized the weaknesses in the banking sector due to excessive leverage and recourse to short term funding as a key risk factor in the recent crisis; see, in addition to Gorton and Andrew (2012), Morris and Shin (2008) and Hanson et al. (2011). The interbank market was a facilitator in the turmoil, not the originating factor.

The real sector of the model consists of a conventional macroeconomic model with nominal rigidities; in a model used for policy analyses, the latter are useful because they help better match the empirical evidence on monetary transmission on output and inflation.

5.1. Households

There is a continuum of identical households who consume, save, work and make portfolio decisions. Households save by lending to financial intermediaries, in the form of demand deposits and bank capital. To allow aggregation within a representative agent framework we assume that in every period a fraction γ of household members are bank capitalists and a fraction $(1 - \gamma)$ are workers/depositors.¹⁶ Hence households also own financial intermediaries. Bank capitalists remain engaged in their business activity next period with a probability θ , independent of history.¹⁷ Workers are employed either in the production sector or in the banking sector, as bank managers; both return their earnings to the household. Bank dividends, earned by bank capitalists who remain in business, are assumed to be passed on to the new bank capitalists and reinvested in the bank (details below in Section 5.2). Households maximize the following discounted sum of utilities:

$$E_0 \sum_{t=0}^{\infty} \beta^t U(C_t, N_t) \quad (1)$$

where C_t denotes aggregate consumption and N_t denotes labor hours. Households save and invest in bank demand deposits and bank capital (as explained above returns on bank capital are reinvested), both entail some risk. Demand deposits, D_t , pay a gross nominal contractual return R_t . Due to the possibility of bank runs, the return on demand deposits is subject to a time-varying risk; the expected return on demand deposits is $R_t(1 - \phi_t g_t)$, where ϕ_t is the probability of run and g_t is the

¹⁴ Our claim is based on data collected from the SIFMA website for the US.

¹⁵ Federal Reserve Board data; see <http://www.federalreserve.gov/releases/H8/default.htm>.

¹⁶ We could alternatively assume two sets of households, one composed solely of risk averse workers and one composed solely of finitely lived and risk neutral bank managers. This alternative assumption would not affect the main channels of the monetary transmission mechanism in our model. The only difference would consist in the addition of a separate consumption function for bank capitalists. Since bank capitalists consist of a small fraction of the population, their consumption would not quantitatively affect the dynamic of the real economy.

¹⁷ This finite survival scheme is needed to avoid that bankers accumulate enough wealth to remove the funding constraint. A fraction $(1 - \theta)$ of bank capitalists exit in every period, becoming workers, and a corresponding fraction of workers become bank capitalists every period, so that the share of bank capitalists, γ , and workers remains constant.

expected loss per unit of deposits conditional on a run, as explained in [Appendix A](#).¹⁸ Households own the production sector, from which they receive nominal profits for an amount, Θ_t . Let T_t be net transfers to the public sector (lump sum taxes, equal to public expenditures). The budget constraint is¹⁹

$$P_t C_t + T_t + D_t \leq W_t N_t + \Theta_t + \Xi_t + R_{t-1}(1 - \phi_{t-1} g_{t-1}) D_{t-1} \quad (2)$$

where W_t is the unitary wage and Ξ_t are the total revenues earned by bank managers. Households choose the set of processes $\{C_t, N_t\}_{t=0}^{\infty}$ and demand deposits $\{D_t\}_{t=0}^{\infty}$, taking as given the set of processes $\{P_t, W_t, R_t\}_{t=0}^{\infty}$ and the initial value of demand deposits D_0 so as to maximize (1) subject to (2). The following optimality conditions hold

$$\frac{W_t}{P_t} = -\frac{U_{n,t}}{U_{c,t}} \quad (3)$$

$$U_{c,t} = \beta E_t \left[\frac{R_t}{\pi_{t+1}} (1 - \phi_t g_t) U_{c,t+1} \right] \quad (4)$$

where $\pi_{t+1} = P_{t+1}/P_t$. Eq. (3) gives the optimal choice for labor supply. Eq. (4) gives the Euler condition with respect to demand deposits. Optimality requires that the first order conditions and no-Ponzi game conditions are simultaneously satisfied.

5.2. Intermediation sector

The intermediation sector collects funds from outside investors (demand depositors, holding demand deposits subject to a service constraint, and bank capitalists) and allocates them to entrepreneurs, who undertake capital investment.²⁰ Total bank funds, L_t , are therefore allocated to finance the total value of capital investment, $Q_t K_{t+1}$ (where K_{t+1} is the aggregate stock of capital investment and Q_t is the re-sell price of the capital good, which will be derived endogenously in Section 5.4.1; occasionally we will refer to Q_t as the asset price or the Tobin's Q). Firms finance investment fully with bank lending. The returns to capital investment has a general aggregate component, represented by the marginal productivity of capital plus the capital gains obtained through the resale market. The return accruing to the intermediary (bank) is subject to an idiosyncratic shock. As already mentioned, the bank manager maximizes the total expected return to both financiers; since funding markets are competitive, this is equivalent to maximizing the bank manager's return, see [Allen and Gale \(2007\)](#). To maintain bank managers' incentives to commit his technological skills depositors can threaten a run. It is assumed that depositors receive precise signals on the projects' returns²¹: when returns are too low, a collective action problem materializes and depositors run the bank. A run entails costly project liquidation, which also produces aggregate resource costs. Outside financiers and bank managers are also linked by a contractual agreement, according to which bank managers receive ex post a share of total expected returns. Linking bank managers' fee to the expected returns through the bargaining agreement helps us to maintain managers' incentive to maximize expected returns. The presence of demand deposits (as opposed to other long terms deposit contracts) avoids the threat of renegotiation: any attempt of the bank manager to renegotiate the contract will set off a run, which by forcing costly liquidation also destroys bank managers' residual claims.²²

Banks are heterogeneous as they run projects whose realization is in general different. However, later on we will show that both the optimal share of demand deposits (and bank capital) and the returns accruing to outside financiers are linear with respect to project value. This allows us to aggregate the equations characterizing the banking sector by simply taking expected values. Based on this and for sake of simplification we omit banks' individual subscripts from the start. Total funds, given by the sum of demand deposits, (D_t), and bank capital, (K_t^B), equal bank lending:

$$L_t = Q_t K_{t+1} = D_t + K_t^B \quad (5)$$

The liability structure of the bank, measured by the deposit share, $d_t = D_t/L_t$,²³ is determined by the bank manager on behalf of the external financiers. The manager sets the bank capital structure so as to maximize the combined expected

¹⁸ Households could in principle invest their savings either lending directly to firms, or by acquiring bank deposits. In the first case, as uninformed investors they would be able to liquidate at most a fraction λ of their investment. As shown in the next section the bank can guarantee to the depositor, in case of run, a payoff at least equal to $(1 + \lambda)(1 - c)(R_t^A - h)/2$. In our benchmark parametrization, the worse case return for the depositor if she invests in the bank is larger than the liquidation value λ , the depositor's outside option. This guarantees the depositor's participation in the contract.

¹⁹ Note that the return from, and the investment in, bank capital do not appear in equation (2), because returns on bank capital are reinvested.

²⁰ To maintain consistency with the hypothesis of a relationship lender, we assume that each bank invests in one project or in a small cluster of projects. The bank manager can indeed acquire information only by monitoring consistently one or a small group of banks. This implies that ex ante we neglect the possibility of full projects' diversification. Notice however that equilibrium runs would materialize even if the bank invests in all projects as long as returns' correlation is different than zero. To maintain tractability we do not consider this case, which would nevertheless be a relevant one.

²¹ Alternatively one could think of depositors forming expectations about banks' returns: those expectations determine expected failure probabilities, thereby being fulfilled in equilibrium. See among others [Kaminsky and Reinhart \(1999\)](#), [Calomiris and Mason \(2003\)](#) for evidence on the links between banks runs and fundamentals.

²² See also [Diamond and Rajan \(2000, 2001\)](#) for a similar logic.

²³ In our simple bank balance sheet the deposit share is the complement to unity of the capital share, $d_t = 1 - \frac{K_t^B}{L_t}$. Hence we have a monotonic positive relation between d_t and the bank's leverage, $\frac{L_t}{K_t^B}$. In the subsequent discussion we often refer to d_t as leverage.

(with respect to the idiosyncratic shock observed ex post by the bank manager) return of depositors and capitalists, in exchange for a fee, set according to the bargaining contractual agreement.

Individual depositors are served sequentially and fully as they come to the bank for withdrawal; bank capitalists are rewarded pro-quota after all depositors are served. This payoff mechanism exposes the bank to runs, that occur when the uncertain return from the project is insufficient to reimburse all depositors. As soon as depositors realize that the payoff is insufficient, they run the bank and force the liquidation of the project; in this case the bank capital holders get zero while depositors get the market value of the liquidated loan.²⁴

The bank asset side yields an expected return R_t^A , homogeneous across banks (the link between the average return and the real economy is detailed below) but subject to an idiosyncratic shock x_t with a uniform distribution defined in the space $\{-h; h\}$.²⁵ As explained above the bank is a relationship lender: by financing the project, it acquires a specialized non-sellable knowledge of its characteristics that determines an advantage in extracting value from it before the project is concluded, relative to other agents. For this reason the bank is able to repossess the entire return $R_t^A + x_t$. If outside investors (depositors or bank capitalists) try to liquidate the project without the assistance of the bank manager, they are able to obtain only a fraction λ of the return. This gives the bank a bargaining power, that allows us to extract a rent, proportional to the remaining part $(1 - \lambda)$. Notice that, since bank capitalists bear the risk of run, the bank manager rewards them in the no run states by assigning them part of the rents, $(1 - \lambda)$.

The timing is as follows. At time t , the bank manager decides the optimal capital structure, expressed by the ratio of demand deposits to the total cost of the project, d_t , and collects the funds. At time $t + 1$, the project's outcome is revealed, the bank manager acquires the return R_t^A , and payments to depositors and capitalists are made. A new round of projects starts.

Even if the full value is extracted from the project, without loss of relationship knowledge, a bank run entails a specific cost $1 > c \geq 0$. When a run occurs, the value of the project loses a constant fraction c , that can be interpreted as arising from early liquidation. Notice that this cost materializes only in the event that a run occurs.

Consider the payoffs to each of our players, namely the depositor, the bank capitalist and the bank manager. Three possible cases arise.

Case A: Run for sure. The return is too low to pay depositors; $R_t^A + x_t < R_t d_t$. Payoffs in case of run are distributed as follows. Capitalists receive the leftover after depositors are served, so they get zero in this case. Depositors, in the absence of bank intervention, would get only a fraction $\lambda(1 - c)(R_t^A + x_t)$ of the project's outcome. The remainder $(1 - \lambda)(1 - c)(R_t^A + x_t)$ is split in half between depositors and the bank manager.²⁶ Therefore, depositors get

$$\frac{(1 + \lambda)(1 - c)(R_t^A + x_t)}{2} \quad (6)$$

and the bank manager gets

$$\frac{(1 - \lambda)(1 - c)(R_t^A + x_t)}{2} \quad (7)$$

Case B: Run only without the bank. The return is high enough to allow depositors to be served if the project's value is extracted by the bank manager, but not otherwise, i.e. $\lambda(R_t^A + x_t) < R_t d_t \leq (R_t^A + x_t)$. In equilibrium the run does not occur, so depositors are paid in full, $R_t d_t$, and the remainder is split in half between the bank manager and the capitalists, each getting $(R_t^A + x_t - R_t d_t)/2$. Total payment to outsiders is $(R_t^A + x_t + R_t d_t)/2$.

Case C: No run for sure. The return is high enough to allow all depositors to be served, with or without the bank's participation. This happens if $R_t d_t \leq \lambda(R_t^A + x_t)$. Depositors get $R_t d_t$. However, unlike in the previous case, now the capitalists have a higher bargaining power because they could decide to liquidate the project alone and pay the depositors in full, getting $\lambda(R_t^A + x_t) - R_t d_t$. This value is thus a lower bound for them. The bank manager can extract $(R_t^A + x_t) - R_t d_t$: once again the surplus arising by the bank intervention is split in half with the bank capitalists. Hence the bank manager gets

$$\left\{ \frac{[(R_t^A + x_t) - R_t d_t]}{2} - \frac{[\lambda(R_t^A + x_t) - R_t d_t]}{2} \right\} = \frac{(1 - \lambda)(R_t^A + x_t)}{2} \quad (8)$$

an amount lower than the one the capitalist gets. Total payment to outsiders is

$$\frac{(1 + \lambda)(R_t^A + x_t)}{2}$$

²⁴ As explained so far bank runs in this model work as discipline devices, hence, as also pointed out in [Diamond and Rajan \(2001\)](#), in this context deposit insurance is inefficient as it distorts banks' incentives.

²⁵ In [Angeloni and Faia \(2013\)](#) we show that results are unchanged also when assuming a logistic or a normal distribution. The uniform distribution is chosen as benchmark as it allows us to work out an analytical solution of the deposit ratio and to gain intuition regarding the main mechanisms.

²⁶ In [Angeloni and Faia \(2013\)](#) we show that different bargaining share between outside investors and bank managers would not affect the results. The equal split is chosen for analytical simplicity.

The manager chooses d_t to maximize the expected payoff to outside investors; summing up the total expected payments to them in the three cases delivers the following expression:

$$\begin{aligned} & \frac{1}{2h} \int_{-h}^{R_t d_t - R_t^A} \frac{(1+\lambda)(1-c)(R_t^A + x_t)}{2} dx_t + \frac{1}{2h} \int_{R_t d_t - R_t^A}^{R_t d_t / \lambda - R_t^A} \frac{(R_t^A + x_t) + R_t d_t}{2} dx_t \\ & + \frac{1}{2h} \int_{R_t d_t / \lambda - R_t^A}^h \frac{(1+\lambda)(R_t^A + x_t)}{2} dx_t \end{aligned} \quad (9)$$

In [Appendix B](#) we show that the value of d_t that maximizes Eq. (9) is comprised in the interval $\lambda(R_t^A + h)/R_t < d_t < (R_t^A + h)/R_t$. In this zone (see region D in our [Appendix B](#)), the third integral in the equation vanishes and the expression reduces to

$$\frac{1}{2h} \int_{-h}^{R_t d_t - R_t^A} \frac{(1+\lambda)(1-c)(R_t^A + x_t)}{2} dx_t + \frac{1}{2h} \int_{R_t d_t - R_t^A}^h \frac{(R_t^A + x_t) + R_t d_t}{2} dx_t \quad (10)$$

The above function is a piece-wise concave function (see graph in [Appendix B](#)), hence the second order condition is satisfied. Differentiating and solving for d_t yields the following equilibrium condition:

$$d_t = z \frac{R_t^A + h}{R_t} \quad (11)$$

where $z = 1/(2 - \lambda + c(1 + \lambda))$. Note that the equilibrium deposit ratio, d_t , is inversely proportional to R_t ; this is straightforward because d_t and R_t appear only in multiplicative form in the outsiders' payoff function (10). Moreover, d_t is directly proportional to $R_t^A + h$, the upper limit of the distribution of payoffs. The intuition can be grasped by inspecting equation (10). At the margin, an increase in the deposit ratio affects the payoff function through two channels. First, by increasing the range of realizations of x where a run occurs (raising the upper limit of the first integral) and decreasing the range where a run does not occur (raising the lower limit of the second integral). This effect does not depend on either R_t^A or h . The second channel is an increase of the payoff to outsiders for each x_t in the interval where a run does not occur, i.e. the interval of the second integral of (10). This effect is proportional to $R_t^A + h - R_t d_t$, the size of this interval. From this we can see that the optimal d_t must be homogeneous of degree one in $R_t^A + h$.²⁷

Note that the sources of deviation from a frictionless Modigliani–Miller world in our model are given by the relationship lender's advantage, $(1 - \lambda)$, and the cost of run, c . If the first is zero ($\lambda = 1$), the bank manager's payoff vanishes and the problem no longer has a closed-form solution given by Eq. (11). If, moreover, $c = 0$, the capital structure of the bank has no effect on the value of the bank: the expected return for its investors, depositors and bank capitalists, equals R_t^A regardless of the value of d_t .

Note also that the parameter z is positively related to λ and negatively related to c . Intuitively, an increase of c (a higher cost of run) decreases the optimal deposit ratio, as does a decrease of λ (a stronger relationship lender effect), for any given value of the bank lending premium $(R_t^A + h)/R_t$.

From Eq. (11) we derive an expression for total bank capital which reads as follows

$$K_t^B = \left(1 - z \frac{R_t^A + h}{R_t}\right) Q_t K_{t+1} \quad (12)$$

The last equation shows that our model also features a traditional banks' balance sheet channel: a fall in the policy rate, by raising asset prices, also helps boost projects and banks' balance sheet values. An increase in the aggregate project value, $Q_t K_{t+1}$, induces banks to increase external finance, both in the form of demand deposits and bank capital. As explained above, following a fall in the policy rate, banks in our model tend to increase the share of demand deposits more than proportionally compared to bank capital. Such a shift will also increase the probability of banks' runs as we show next.

Finally, our model allows us to compute the probability of occurrence of bank runs which is defined as follows:

$$\phi_t = \frac{1}{2h} \int_{-h}^{R_t d_t - R_t^A} dx_t = \frac{1}{2} \left(1 - \frac{R_t^A - R_t d_t}{h}\right) \quad (13)$$

We will refer to ϕ_t also as bank riskiness.

²⁷ More formally, a marginal increase in the deposit ratio increases the range of x_t where a run occurs, by raising the upper limit of the first integral; this effect increases the overall payoff to outsiders by $(1/2h)((1+\lambda)(1-c)/2)R_t d_t$. A marginal increase in the deposit ratio also decreases the range of x_t where a run does not occur, by raising the lower limit of the second integral; the effect of this on the payoff is negative and equal to $-(1/2h)R_t^2 d_t$. Moreover, it also increases the return to outsiders for each value of x_t where a run does not occur; this effect is $(1/2h) \left(\int_{R_t d_t - R_t^A}^h \frac{1}{2} dx_t \right) R_t = (1/2h) \left((R_t^A + h - R_t d_t)/2 \right) R_t$. Equating to zero the sum of these effects and solving for d_t yield Eq. (11).

Eqs. (11) and (13) capture the essence of the risk taking channel in our model. As we will show in Appendix C, a contractionary monetary policy shock that raises R_t also increases R_t^A . Replacing Eq. (11) in Eq. (13) results in

$$\phi_t = \frac{1}{2} \left(1 - \frac{R_t^A(1-z) - zh}{h} \right) \quad (14)$$

Since $z < 1$ (as long as $0 < \lambda < 1$ and $0 < c < 1$), Eq. (14) immediately shows that an increase in R_t^A reduces bank risk ϕ_t . One can show that also leverage, d_t , decreases for plausible parameter values.

More intuitively, a fall in the policy rate lowers the cost of short term funding. This induces the bank manager to shift toward short term funding as opposed to bank capital, which instead comes along with the additional rents extractions. The bank managers must balance the benefits of cheaper external funding with the costs of an increase in bank riskiness: on balance the share of short term lending increases, leading ex post to higher risks of bank runs. Although the bank manager acts optimally from an individual point of view, higher probability of bank runs has ex post social resource costs, given by the expected losses ensuing projects' liquidation: atomistic bankers do not internalize such social costs, thereby they leverage more than it would be optimal.

5.3. Bank capital accumulation

After remunerating depositors and paying the fee to the manager, a return accrues to the bank capitalist as retained earning. Bank capitalists who remain in business accumulate all their returns. Bank capital accumulates from retained earnings as follows (again individual subscripts are omitted since aggregation does not change the shape of the aggregate bank capital accumulation):²⁸

$$K_t^B = \frac{\theta}{\pi_t} [K_{t-1}^B + R_t^{BK} L_t] \quad (15)$$

where R_t^{BK} is the unitary return to the capitalist and $\pi_t = P_t/P_{t-1}$ is the inflation, which will be defined and derived in this section and which enters here since the accumulation involves bank capital at different dates. The parameter θ is the bank survival rate. R_t^{BK} can be derived from Eq. (10) as follows:

$$R_t^{BK} = \frac{1}{2h} \int_{R_t d_t - R_t^A}^h \frac{(R_t^A + x_t) - R_t d_t}{2} dx_t = \frac{(R_t^A + h - R_t d_t)^2}{8h} \quad (16)$$

Note that this expression considers only the no-run state because if a run occurs the capitalist receives no return. The accumulation of bank capital is obtained substituting (16) into (15):

$$K_t^B = \frac{\theta}{\pi_t} \left[K_{t-1}^B + \frac{(R_t^A + h - R_t d_t)^2}{8h} L_t \right] \quad (17)$$

The bank capital structure depends on several counterbalancing factors. One can interpret Eq. (12) as a “demand” for bank capital given the volume of loans L_t and the interest rate structure (R_t , R_t^A), while Eq. (17) can be seen as a “supply” of bank capital in the following period.

5.4. Intermediate good producers

Given that our focus is on the analysis of the monetary transmission mechanism, we also allow for non-neutral effects of monetary policy; to that aim we introduce nominal rigidities, by assuming quadratic adjustment costs on prices. Final goods in this economy are obtained by assembling, through a conventional Dixit Stiglitz aggregator, intermediate goods. Each firm i in the intermediate good sector has monopolistic power in the production of its own variety and therefore has leverage in setting the price. In changing prices it faces a quadratic cost equal to $(\vartheta/2)(P_t(i)/P_{t-1}(i) - 1)^2$, where the parameter ϑ measures the degree of nominal price rigidity. The higher the ϑ the more sluggish is the adjustment of nominal prices. Each firm assembles labor (supplied by the workers) and (finished) entrepreneurial capital to operate a constant return to scale production function for the variety i of the intermediate good: $Y_t(i) = A_t F(N_t(i), K_t(i))$. Each monopolistic firm chooses a sequence $\{K_t(i), N_t(i), P_t(i)\}$, taking nominal wage rates W_t and the rental rate of capital Z_t , as given, in order to maximize expected discounted nominal profits:

$$E_0 \left\{ \sum_{t=0}^{\infty} \Lambda_{0,t} \left[P_t(i) Y_t(i) - (W_t N_t(i) + Z_t K_t(i)) - \frac{\vartheta}{2} \left[\frac{P_t(i)}{P_{t-1}(i)} - \pi \right]^2 P_t \right] \right\} \quad (18)$$

²⁸ We assume that bank capitalists who exit business in every period transfer their wealth to capitalists who remain in business. Hence the aggregate wealth also includes an additional term (which, to facilitate notation, we do not report in the equations of the main text): $\Sigma_t = \gamma K_{t-1}^B$. This term is parametrized so that bank net worth never falls below zero.

subject to the following aggregate demand constraint $A_t F_t(\bullet) \leq Y_t(i) = (P_t(i)/P_t)^{-\varepsilon} Y_t$, where $\Lambda_{0,t} = U_{c,t+1}/U_{c,t}$ is the households' stochastic discount factor as obtained from the Euler condition (4).

Let us denote by $\{mc_t\}_{t=0}^\infty$ the sequence of Lagrange multipliers on the above demand constraint and by $\tilde{p}_t \equiv P_t(i)/P_t$ the relative price of variety i . After dividing the profit function by the aggregate price P_t and taking first order conditions, we obtain

$$\frac{W_t}{P_t} = mc_t A_t F_{n,t}; \quad \frac{Z_t}{P_t} = mc_t A_t F_{k,t} \quad (19)$$

$$0 = U_{c,t} Y_t \tilde{p}_t^{-\varepsilon} ((1-\varepsilon) + \varepsilon mc_t) - \vartheta \left[\pi_t \frac{\tilde{p}_t}{\tilde{p}_{t-1}} - 1 \right] \frac{\pi_t}{\tilde{p}_{t-1}} U_{c,t} \\ + \vartheta E_t \left\{ \left[\pi_{t+1} \frac{\tilde{p}_{t+1}}{\tilde{p}_t} - 1 \right] U_{c,t+1} \pi_{t+1} \frac{\tilde{p}_{t+1}}{\tilde{p}_t^2} \right\} \quad (20)$$

where $F_{n,t}$ is the marginal product of labor, $F_{k,t}$ is the marginal product of capital and $\pi_t = P_t/P_{t-1}$ is the gross aggregate inflation rate. Notice that all firms employ an identical capital/labor ratio in equilibrium, so individual prices are all equal in equilibrium. The Lagrange multiplier mc_t plays the role of the real marginal cost of production. In a symmetric equilibrium $\tilde{p}_t = 1$. After substituting the stochastic discount factor, and the condition for a symmetric equilibrium, Eq. (20) takes the following form:

$$U_{c,t}(\pi_t - 1)\pi_t = \beta E_t \{ U_{c,t+1}(\pi_{t+1} - 1)\pi_{t+1} \} + U_{c,t} A_t F_t(\bullet) \frac{\varepsilon}{\vartheta} \left(mc_t - \frac{\varepsilon - 1}{\varepsilon} \right) \quad (21)$$

The above equation is a non-linear forward looking New-Keynesian Phillips curve in which deviations of the real marginal cost from its desired steady state value are the driving force of inflation.

Using the equation for labor supply, (3), and for labor demand, (19), we can derive at this stage also the labor market equilibrium condition, which reads as follows:

$$-\frac{U_{n,t}}{U_{c,t}} = mc_t A_t F_{n,t} \quad (22)$$

5.4.1. Capital producers

Investment decisions are taken by a sector of capital producers that face adjustment costs: the latter are introduced to obtain a time-varying price of capital, namely a conventional Tobin's Q . A competitive sector of capital producers combines investment, expressed in the same composite index as the final good, hence with price P_t , and existing capital stock to produce new capital goods. This activity entails physical adjustment costs. The corresponding constant-returns-to-scale production function is $\chi(I_t/K_t)K_t$, so that capital accumulation obeys

$$K_{t+1} = (1-\delta)K_t + \chi\left(\frac{I_t}{K_t}\right)K_t \quad (23)$$

where $\chi(\bullet)$ is increasing and convex. Define Q_t as the re-sell price of the capital good. Capital producers maximize profits $Q_t \chi(I_t/K_t)K_t - P_t I_t$, implying the following optimal price of assets:

$$Q_t \chi\left(\frac{I_t}{K_t}\right) = P_t \quad (24)$$

The gross (nominal) return from holding one unit of capital between t and $t+1$ is composed of the rental rate plus the re-sell price of capital (net of depreciation and physical adjustment costs):

$$Y_t^k \equiv Z_t + Q_t \left((1-\delta) - \chi'\left(\frac{I_t}{K_t}\right) \frac{I_t}{K_t} + \chi\left(\frac{I_t}{K_t}\right) \right) \quad (25)$$

The gross (real) return to entrepreneurs from holding a unit of capital between t and $t+1$ is equalized in equilibrium to the gross (real) return that entrepreneurs return to banks for their loan services, R_{t+1}^A :

$$\frac{R_{t+1}^A}{\pi_{t+1}} \equiv \frac{Y_{t+1}^A}{Q_t} = \frac{mc_{t+1} A_{t+1} F_{k,t+1} + Q_{t+1} \left((1-\delta) - \chi'\left(\frac{I_{t+1}}{K_{t+1}}\right) \frac{I_{t+1}}{K_{t+1}} + \phi\left(\frac{I_{t+1}}{K_{t+1}}\right) \right)}{Q_t} \quad (26)$$

Eq. (26) establishes that the aggregate return to capital must equate the marginal productivity of capital, $mc_{t+1} A_{t+1} F_{k,t+1}$, plus the capital gains, Q_{t+1}/Q_t , obtained by reselling capital at the end of period t . The capital sold at the end of period t is net of depreciation and of the adjustment costs to investment.

5.5. Official sector and market clearing

We assume that monetary policy is conducted by means of an interest rate reaction function of this form

$$\ln\left(\frac{1+R_t}{1+\bar{R}}\right) = \left[\phi_\pi \ln\left(\frac{\pi_t}{\bar{\pi}}\right) + \phi_y \ln\left(\frac{Y_t}{\bar{Y}}\right)\right] + m_t \quad (27)$$

All variables at the denominator, without time subscript, are the target or steady state. The variable m_t is a monetary policy shock whose process is described in the calibration section. Parameters in the monetary policy rule have been calibrated according to the standard Taylor rule, namely $\phi_\pi = 1.5$ and $\phi_y = 0.5/4$.

The government runs a balanced budget and uses lump sum taxation to finance exogenous government expenditure, hence $T_t = G_t$.

Equilibrium in the final goods market requires that the production of the final good equals private consumption, investment, public spending, and the various resource costs. The combined resource constraints, inclusive of government budget, read as follows:

$$Y_t - \mathcal{L}_t = C_t + I_t + G_t + \frac{\theta}{2}(\pi_t - 1)^2 \quad (28)$$

In the above equation, G_t is the government consumption of the final good which evolves exogenously (see calibration section) and is assumed to be financed by lump sum taxes. The term $(\theta/2)(\pi_t - 1)^2$ represents the aggregate costs associated with the price adjustment process. The term $\mathcal{L}_t = (1/2h) \int_{-h}^{R_t d_t - R_t^A} c R_t^A Q_t K_{t+1} dx_t$ represents the expected cost of project liquidation in the event of a run; it corresponds to the society's resource loss due to bank risk, in expected terms.

5.5.1. Definition of competitive equilibria

Definition. For a given sequence of nominal interest rate $\{R_t\}_{t=0}^\infty$, for given initial conditions on asset evolution $\{K_0, D_0, K_0^B\}_{t=0}^\infty$ and for a given set of exogenous processes $\{A_t, G_t, m_t\}_{t=0}^\infty$ a determinate competitive equilibrium for this economy is a sequence of allocations and prices $\{C_t, N_t, d_t, K_t^B, I_t, K_{t+1}, Y_t, \pi_t, mc_t, Q_t, R_t^A\}_{t=0}^\infty$ which satisfy Eqs. (4), (22), (11), (21), (24), (26), (17), (23), (28), (5) and $Y_t = A_t F(N_t, K_t)$.

The equations above summarize the equilibrium conditions for our economy. Eqs. (4) and (22) are the optimality conditions for the consumer's optimization problem, Eq. (11) is the optimality condition for the bank's optimization problem, Eqs. (21), (24), and (26) solve the firms' optimization problem, Eqs. (17) and (23) are the wealth accumulation equations and Eqs. (28) and (5) are the technological constraints.

In the quantitative simulations the model is solved in first-order approximation when discussing the impulse response functions and in second order approximations when discussing the results on asset price and output volatilities. The first order approximation allows us to compare the impulse response functions with the VAR equivalent, which are linear. The second order approximations allow us to take into account the effects of model non-linearities on the asset price volatilities. In both cases the model is approximated around the stochastic steady state characterized by the long run distribution of bank projects' idiosyncratic returns: details on the calibration of the probability distribution are given in the calibration section.

5.6. Transmission channels in our model: balance sheet and risk taking

It is useful at this stage to briefly discuss the monetary policy transmission channels in our model. Our model features both the traditional balance sheet channel and a risk taking channel on the funding side. We already explained the distinction between the two in Section 4; let us see this now in more detail and in relation to our model.

As well known, the balance sheet channel induces an amplification of business cycle fluctuations relative to a standard macroeconomic model without credit frictions. In our model a contractionary monetary policy reduces asset prices, Q_t as output and investment fall, the asset price falls according to Eq. (24). As a result, the size of the bank balance sheet declines; it turns out that both bank capital and deposit fall. The credit contraction carries over to the following periods. From Eq. (17) indeed we see that next period's bank capital value is reduced, implying a shrinking in next period lending as from equation $L_{t+1} = D_{t+1} + K_{t+1}^B$. The ensuing fall in investment triggers further subsequent falls in asset prices and may generate a progressive negative spiral, akin to a fire sale. Due to this channel alone, the recessionary effects of a monetary policy contraction would be amplified. Also notice that in the traditional literature on the balance sheet channel falls in the value of banks' balance sheet would increase their risk of default.

However, our banking sector features in addition a risk taking channel on the funding side. A contractionary policy, by increasing the cost of short term funding reduces banks' leverage, d_t , and the probability of bank runs. The fall in bank risk, ϕ_t , also reduces the resource costs of projects' liquidation: this dampens the recessionary effects of monetary contractions.

Both the balance sheet and the risk taking channel imply a fall in short term liabilities following a monetary contraction. However the risk taking channel generates a fall in short term liabilities which is larger relatively to the fall in bank capital. By reducing banks' leverage relatively more the monetary contraction, associated with a risk taking channel, tends to dampen banks' risk. The reduction in banks' risk also dampens fluctuations in the resource costs associated with it.

The numerical simulations of our model (as reported below) show overall a dampened business cycle and a fall in leverage with monetary contractions; thereby they signal that the risk taking channel is predominant.

5.7. Parameter values

Household preferences and production: The time unit is the quarter. The utility function of households is $U(C_t, N_t) = (C_t^{1-\sigma} - 1)/(1-\sigma) + \nu \log(1 - N_t)$, with $\sigma = 1$, as in most real business cycle literature. We set ν set equal to 3, chosen in such a way to generate a steady-state level of employment $N \approx 0.3$. We set the discount factor $\beta = 0.99$, so that the annual real interest rate is equal to 4%. We assume a Cobb–Douglas production function $F(\bullet) = K_t^\alpha (N_t)^{1-\alpha}$, with $\alpha = 0.3$. The quarterly aggregate capital depreciation rate δ is 0.025, the elasticity of substitution between varieties 6. The adjustment cost on capital takes the following form: $[(\chi/2)(I_t/K_t - \delta)^2 K_t]$ and the parameter χ is set so that the volatility of investment is larger than the volatility of output, consistently with empirical evidence: this implies an elasticity of asset prices to investment of 2.

In order to parametrize the degree of price stickiness ϑ , we rely on the comparison between the slope of the log-linear Phillips curve in our model, $(\varepsilon - 1)/\vartheta$, with that arising under a Calvo–Yun setup, which is given by $(1 - \hat{\vartheta})(1 - \beta\hat{\vartheta})/\hat{\vartheta}$, where $\hat{\vartheta}$ is the probability of not resetting the price in any given period. Given the values for the demand elasticity $\varepsilon = 6$, a value of $\hat{\vartheta} = 0.75$, which is compatible with most empirical evidence, the comparison delivers a value for the price stickiness parameter in our model of $\vartheta = Y\hat{\vartheta}(\varepsilon - 1)/(1 - \hat{\vartheta})(1 - \beta\hat{\vartheta}) \approx 30$, where Y is the steady-state output.

Banks: To calibrate h we have calculated the average volatility of bank stocks over the last 10 years (GARCH estimates and realized volatilities yield roughly the same result) which is somewhat below 0.3, and multiply this by the square root of 3, the ratio of the maximum deviation to the standard deviation of a uniform distribution. We take 0.4 as our benchmark.

One way to interpret λ is to see it as the ratio of two present values of the project, the first at the interest rate applied to firms' external finance, the second discounted at the bank internal finance rate (the money market rate). A benchmark estimate can be obtained by taking the historical pre-crisis values of the money market rate and the bank lending rate. In the US over the last 20 years, based on 30-year mortgage loans, the spread has been around 3%. This leads to a value of λ around 0.5. In the numerical simulations we have chosen a value of 0.45. We parametrize the survival rate of banks, θ , at 0.97, a value compatible with an average horizon of 10 years. Notice that the parameter $(1 - \theta)$ is meant to capture only the exogenous exit rates, not the failure rates. Finally, we use a benchmark value of the social cost of a bank run, c , of 0.1, equal to the direct costs of resolution estimated by James (1991) on a sample of banks liquidated by the FDIC.

Shocks: There are three macro-shocks in the model. The first, a productivity shock, is simulated in order to describe the transmission mechanism at work in our model. The monetary policy shock is simulated to analyze the risk taking channel. Total factor productivity is calibrated according to standard RBC processes: it evolves as an AR(1) of the following form: $A_t = A_{t-1}^\rho \exp(\varepsilon_t^a)$, where the steady-state value A is normalized to unity, $\rho_a = 0.95$ and where ε_t^a is an i.i.d. shock with standard deviation $\sigma_a = 0.008$. We then have an additive disturbance to the interest rate set through the monetary policy rule. The monetary policy shock is assumed to be moderately persistent (coefficient 0.2), as argued by Rudebusch (2002). Based on the evidence presented in Section 3, and consistently with other empirical results for US and Europe, the standard deviations of the shocks are set to 0.006. Finally, log-government consumption evolves according to the following exogenous process, $\ln(g_t/g) = \rho_g \ln(g_{t-1}/g) + \varepsilon_t^g$, where the steady-state share of government consumption, g , is set so that $g/y = 0.2$ and ε_t^g is an i.i.d. shock with standard deviation σ_g . In accordance with macro-evidence for both the U.S. and Europe, we set $\sigma_g = 0.007$ and $\rho_g = 0.9$.

6. Model analysis and results

We analyze our model along two dimensions. First, we verify, by examining its impulse response functions,²⁹ whether our model reproduces the empirical evidence we presented earlier. Second, to complete the assessment of the relationship between risk, monetary policy and macro-transmission and performance, we analyze the effect of an increase in the volatility of projects' idiosyncratic shocks (h , the investment projects' risks in our model) on the long run level of bank riskiness and output and on the volatility of asset prices and bank returns. As explained below, our model is able to replicate the relations that characterize those variables in the data and in the past literature, but through a novel channel.

To begin with, to introduce the reader to the functioning of the model, Fig. 2 shows impulse responses to a persistent 1% productivity increase.

As expected output raises and inflation falls on impact, due to nominal rigidities. These are standard results common to most New-Keynesian-type models. The ensuing fall in the policy rate, which is set according to a Taylor rule, triggers an increase in the deposit ratio and in bank riskiness, as per Eq. (11). This happens for two reasons. First, the increase in asset

²⁹ The figures show impulse response functions obtained through first-order approximation of the model. This choice is motivated by the need to provide impulse responses which are consistent with those in the VAR, which is linear. Importantly, due to the endogenous nature of our bank risk, a risk taking channel materializes in our model as first order effect of a decrease in the nominal interest rate. On the contrary, the volatilities presented later in the paper are computed with second-order approximations to take into account the effect of nonlinearities and the full cost of risk in our model.

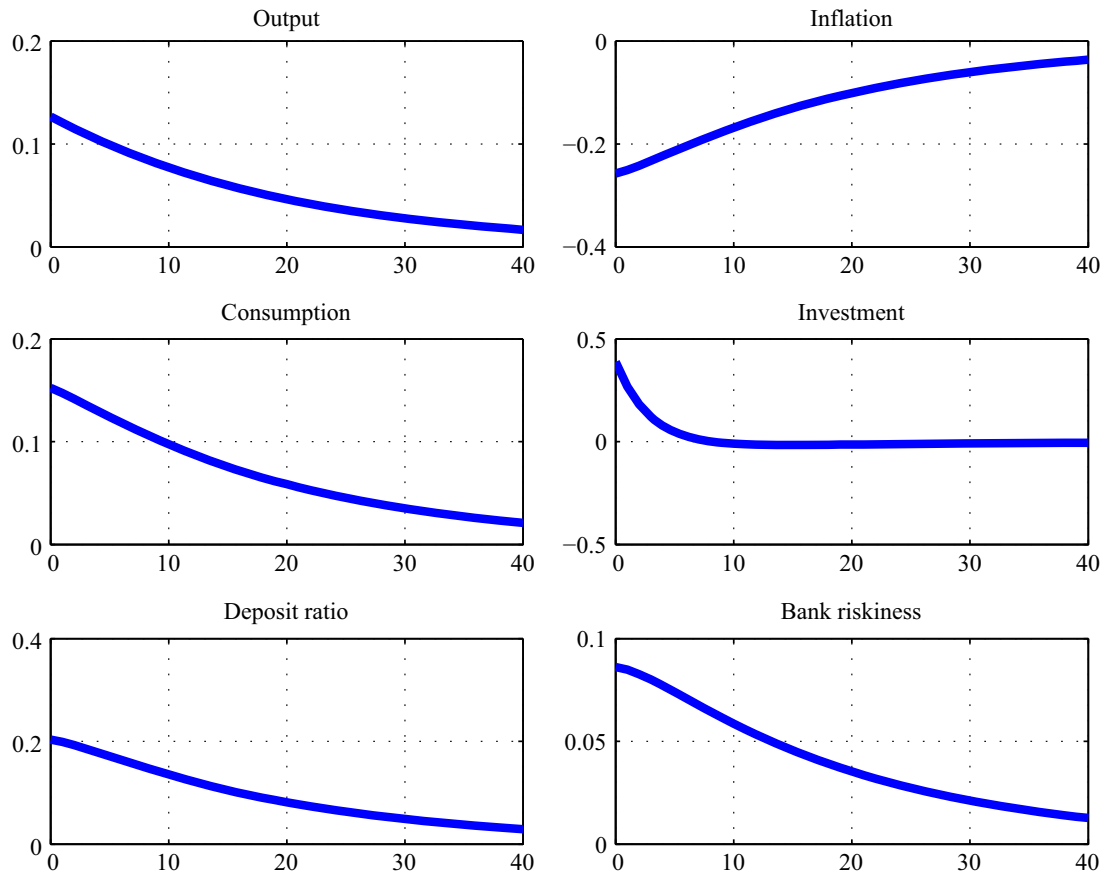


Fig. 2. Impulse response to a positive productivity shock.

prices raises investment and the demand for bank loans. As a consequences banks require higher external funding, that can be provided through demand deposits and/or bank capital. The fall in the nominal interest rate also implies that demand deposits become a cheaper form of external finance, hence bank managers increase the fraction of lending financed by demand deposits. The ensuing increase in bank leverage comes along with an increase in the size of the run region and the probability of bank runs.

We now examine the transmission of a contractionary monetary policy shock. Fig. 3 shows impulse responses to a 1% short term interest shock; solid lines (blue) show our benchmark model with banking. As expected, output, investment and asset prices decline on impact. Due to nominal rigidities, aggregate demand falls. An increase in the policy rate reduces asset prices; this, by reducing the value of banks' assets, also induces a credit squeeze and a fall in investment (balance sheet channel). The risk taking channel on the funding side works as follows. The fall in asset prices and investment triggers a fall of bank funding; this induces a fall in both D_t and K_t^B . As demand deposits are now a relatively more expensive form of funding, the deposit ratio falls by more.³⁰ The fall in demand deposits comes along with a fall in banks' risk, namely the probability of banks' runs.

To better highlight the mechanisms at work in our model we compare these results with those obtained with a standard dynamic New-Keynesian model without banks – dashed (green) lines, labelled DNK. The comparison reveals that in our model the short term impact of a monetary policy contraction is dampened. Bank risk in our model is contractionary; hence, a monetary restriction, due to a decrease of bank risk, reduces output less than would be the case in the absence of a risk taking channel. The fall in bank risk reduces the resource costs Ω , hence inducing an increase in total resources that tends to increase household consumption.³¹

³⁰ When the interest rate, R_t , raises, the return on projects, R_t^A , also raises. The deposit ratio is given by $d_t = z(R_t^A + h)/R_t$. Any increase in R_t^A is dampened by the factor, h . As a result the ratio $(R_t^A + h)/R_t$ falls and the deposit ratio falls.

³¹ The dampening result is reminiscent of analyses showing that relationship lending tends to dampen the impact of monetary shocks on borrowers (e.g. Petersen and Rajan, 1994).

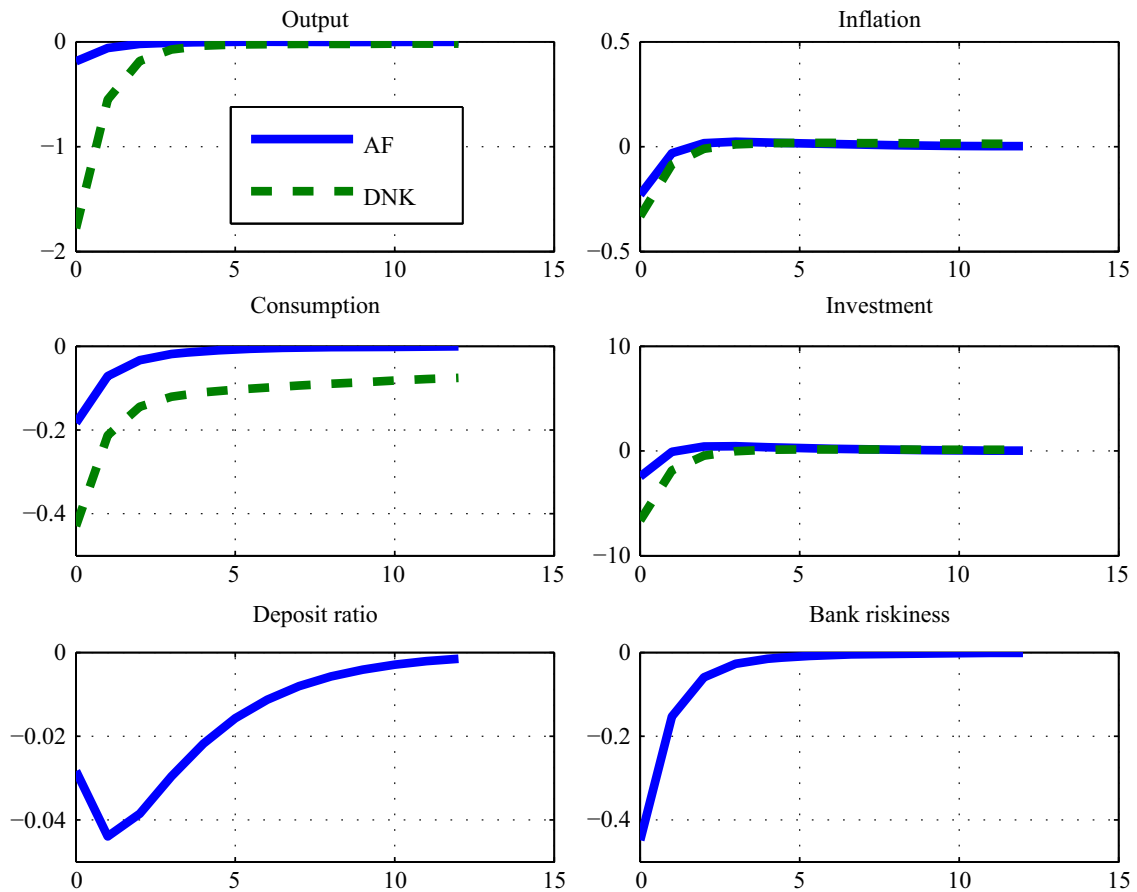


Fig. 3. Impulse response to a monetary restriction (two models).

6.1. The link between risk, financial market volatility and the macroeconomy

Our model sheds new light on the widely explored links between risk on the one side and financial performance (as summarized by volatility and long run level of asset prices) as well as macroeconomic performance (as summarized by volatility and long run level of output) on the other. Several papers have discussed the effect of an increase in asset risk (triggered by the arrival of “bad news”) on the volatility and the long run level of asset prices and/or output. Generally speaking the literature finds that an increase in asset risk triggers an increase in the volatility of both asset price and output and a fall in their long run levels. These links have undergone much greater scrutiny after the financial crisis. The classic result of [Campbell and Hentschel \(1992\)](#), that an increase in stock market volatility, induced by an increase in investment risk, is associated with higher returns and lower stock prices in equilibrium, has been re-examined recently, among others, by [Bae et al. \(2007\)](#) and [Bloom \(2009\)](#). The first paper tries to identify causality, looking at whether it is asset risk, which by raising asset price volatility causes asset prices to decline (as also suggested by Campbell and Hentschel), or else it is the low level of stock prices that, by increasing leverage, drives stock market volatility up. In addition, [Bloom \(2009\)](#) shows that an increase in financial risk increases output volatility and reduces its long run level. We re-examine those links within our model which features endogenous risk formation.

One appealing feature of our model is that we can distinguish between asset or projects' risk (which is captured by the volatility of shocks to projects' returns, h) and endogenous formation of bank risk (probability of bank runs). When projects' risk rises, depositors adjust their run region: such an adjustment process affects the availability of funding to bank, which in turn affects the availability of credit to the economy as well as the long run level and the dynamics of investment and output. To this purpose we examine the effect of an increase in the risk of projects returns (as captured by the idiosyncratic volatility h) on the long run levels and the business cycle volatilities for some variables, computed using second order approximations of the full model to account for first and second order effects of risk in our model.³²

³² To compute volatilities we considered the set of shocks described in the calibration section.

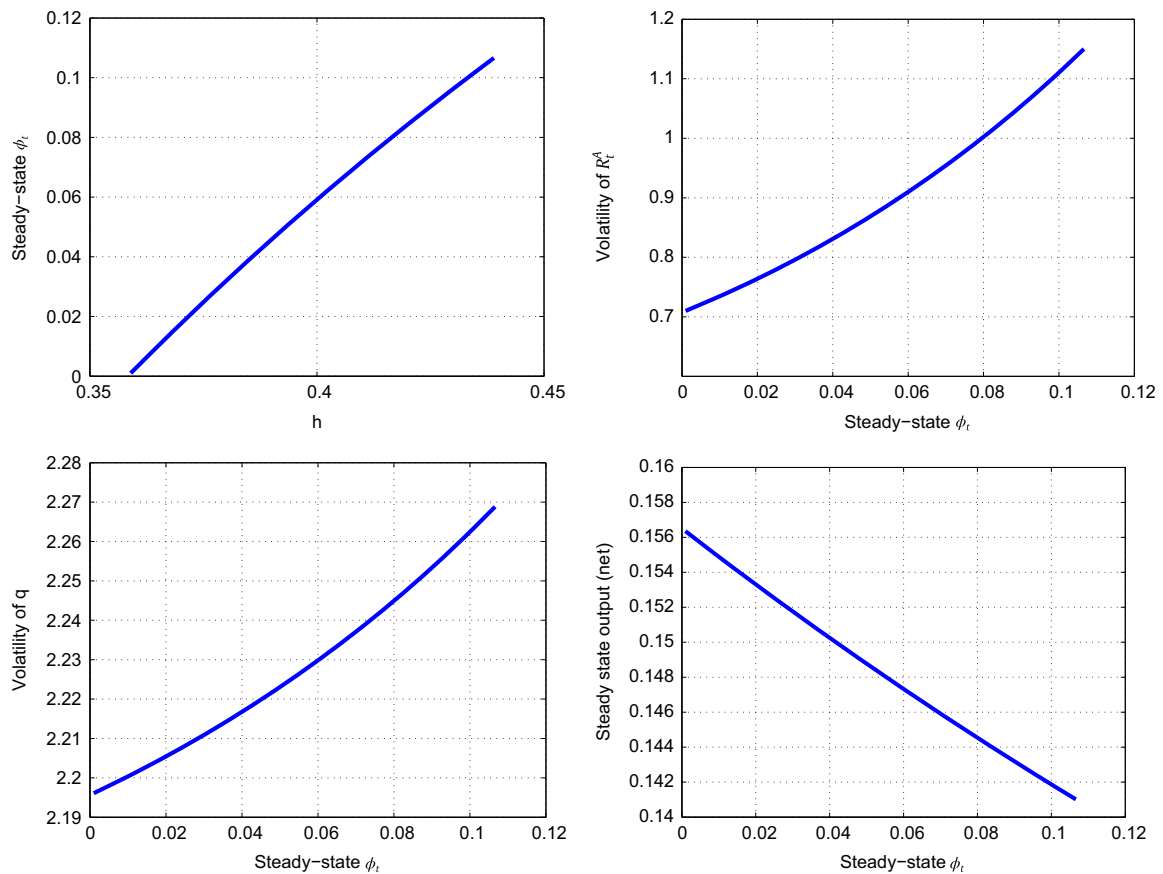


Fig. 4. Mapping among project risk (h), bank risk, volatility of R_t^A , volatility of the asset price and steady state output.

Fig. 4 shows long run levels of bank riskiness and output and the volatilities of asset returns (R_t^A in our model) and asset prices, Q_t . We see first, in panel 1, that an increase in h raises bank risk in the long run (probability of runs). This happens for two reasons. There is first a direct effect. As the probability of extreme events raises, the runs region widens (see Eq. (13)). Second an increase in h induces an increase in bank leverage, see Eq. (11): as the bank is more exposed to demand deposits, the probability of a run increases.

In the long run, higher risk is rewarded with higher return R^A (the steady state value of R_t^A increases). The higher cost of funding induces entrepreneurs to reduce the demand of funding, hence investment in the long run. This coupled with the increase in the log run resource costs of bank risk, Ω , reduces the long run level of output – panel 4 (expressed as percentage output loss relative to the case in which bank risks are zero).

Let us now examine the effects of such shift in risk on business cycle volatilities. To meet the higher level of long run returns, banks' funding and firms' credit availability shall increase by more in response to risk-increasing shocks. This amplified response translates into higher volatility of bank asset returns and asset prices, panels 2 and 3 (the values of these volatility are congruent with the data). This is in turn associated with higher volatility of output, investment and inflation (not shown).³³

These results confirm links already noted in past literature, but also highlight a new channel that stems from the endogenous formation of risk. When investment project risk increases, more bank runs materialize. This destabilizes bank funding and investment (raising their volatility), and reduces output potential in the long run.

All together, these results can help interpret certain developments in the years prior to the crisis. A sequence of positive productivity shocks, alongside with expansionary monetary policy, increased bank leverage. The implication for bank risk was not appreciated immediately by market participants, as witnessed by the fact that credit spreads and ratings remained

³³ For low levels of bank risk, the volatility of output first declines before rising, as bank risk increases. This concave shape is due to the fact that higher bank risk reduces the volatility of the interest rate *net of bank risks*, which is the return relevant for consumer decisions in our model. Hence consumption volatility initially declines before rising. Instead, investment volatility rises monotonically in the whole range.

very favorable for a long period during the leverage buildup. The impact of the monetary expansion on output was positive. But in the end, when risks built up in the economy and became entrenched, they manifested themselves in the form of high risk spreads, high (downward) volatility of output and inflation. The model predicts, in addition, lower steady state output and investment.

7. Conclusions

As a consequence of the financial crisis, a broad reflection is underway on the working of the transmission mechanism of monetary policy in the presence of financial risks. There is a growing perception that existing macroeconomic model that do not incorporate financial sectors and financial risks cannot provide a convincing representation of the effects of monetary policy, particularly when the banking and financial sectors are distressed.

We present new evidence linking monetary policy and bank riskiness through a risk taking channel: lowering policy rates raises bank riskiness, particularly on the funding side. We propose a model with *bank* runs and banks' risk taking that reproduces the main channels highlighted in the time series evidence. Overall, we highlight a new dimension of the monetary policy transmission that calls reflection upon the long run unintended consequences of protracted policy expansions and opens the avenue to a reconsideration of the optimal policy design in presence of financial risk.

Acknowledgments

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Appendix A. Expected loss on risky demand deposits

When the probability of bank run is non-zero, the expected payoff on demand deposits is below the risk-less return R_t . Consider the payoff of demand deposits per unit of funds intermediated by the bank in two events: run for sure and no run (all other cases). In the first case the payoff is $(1+\lambda)(1-c)(R_t^A + x_t)/2$. This holds in the interval of x_t comprised between $[-h; (R_t d_t - R_t^A)]$. The expected value of this payoff is $(1/2h) \int_{-h}^{R_t d_t - R_t^A} ((1+\lambda)(1-c)(R_t^A + x)/2) dx_t$. This can be written, solving the integral and using the expression for the probability of run ϕ_t , Eq. (13), as

$$\begin{aligned} \frac{(1+\lambda)(1-c)}{2} \int_{-h}^{R_t d_t - R_t^A} \frac{(R_t^A + x_t)}{2h} dx_t &= \frac{(1+\lambda)(1-c)}{2} \left[\phi_t R_t^A + \frac{1}{2h} \frac{(R_t d_t - R_t^A)^2 - h^2}{2} \right] \\ &= \phi_t \frac{(1+\lambda)(1-c)}{2} \left(R_t^A + \frac{R_t d_t - R_t^A - h}{2} \right) \\ &= \frac{1}{4} \phi_t (1+\lambda)(1-c) (R_t d_t + R_t^A - h) \end{aligned}$$

In the range of x_t in which the run does not occur, the payoff is equal to $R_t d_t$; its expected value is obtained multiplying it by the probability of the respective event, $(1 - \phi_t)$.

Overall, the expected payoff on demand deposits per unit of intermediated funds therefore is given by

$$\frac{1}{4} \phi_t (1+\lambda)(1-c) (R_t d_t + R_t^A - h) + (1 - \phi_t) R_t d_t$$

The expected loss on demand deposits, relative to the no-default state, per unit of intermediated funds, is obtained by subtracting the above expression from $R_t d_t$, the contractual payoff:

$$R_t d_t - \left[\frac{1}{4} \phi_t (1+\lambda)(1-c) (R_t d_t + R_t^A - h) + (1 - \phi_t) R_t d_t \right]$$

One can also calculate the expected return on demand deposits, i.e. the payoff per unit of demand deposits. This is equal to $R_t(1 - \phi_t g_t)$, where $g_t = \frac{1}{4}(1 + \lambda)(1 - c)(R_t + R_t^A - h)/d_t$.

Appendix B. Optimal deposit ratio

In order to show that the value of d_t that maximizes the function (9) is equal to $(1/R_t)R_t^A + h/(2 - \lambda + c(1 + \lambda))$, we divide the d_t space as follows:

- Interval A: $R_t d_t < \lambda(R_t^A - h)$.
- Interval B: $\lambda(R_t^A - h) < R_t d_t < R_t^A - h$.
- Interval C: $R_t^A - h < R_t d_t < \lambda(R_t^A + h)$.
- Interval D: $\lambda(R_t^A + h) < R_t d_t < R_t^A + h$.
- Interval E: $R_t^A + h < R_t d_t$.

We now analyze the function in each interval, in the following order: A, B, C, E, D. The last one is where we will show the global maximum to be located.

- Interval A: $R_t d_t < \lambda(R_t^A - h)$. The function reduces to $(1/2h) \int_{-h}^h (1 + \lambda)(R_t^A + x_t)/2 dx_t$. This is independent of d_t , hence the function is flat and its level is equal to $\frac{1}{2}R_t^A(1 + \lambda)$.
- Interval B: $\lambda(R_t^A - h) < R_t d_t < R_t^A - h$. The function reduces to

$$\frac{1}{2h} \int_{-h}^{R_t d_t / \lambda - R_t^A} \frac{(R_t^A + x_t) + R_t d_t}{2} dx_t + \frac{1}{2h} \int_{R_t d_t / \lambda - R_t^A}^h \frac{(1 + \lambda)(R_t^A + x_t)}{2} dx_t$$

The first derivative is $(R_t/4h)[(R_t/\lambda)d_t - (R_t^A - h)]$ and the second derivative is $[(1/4h\lambda)R_t^2]$, both positive for all admissible parameter values. Hence in this interval the function is upward sloping and convex.

- Interval C: $R_t^A - h < R_t d_t < \lambda(R_t^A + h)$. The function is equal to

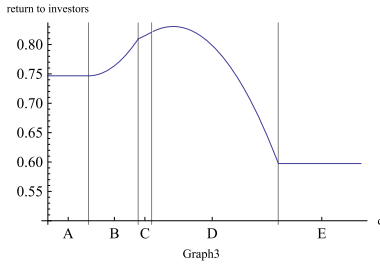
$$\frac{1}{2h} \int_{-h}^{R_t d_t - R_t^A} \frac{(1 + \lambda)(1 - c)(R_t^A + x_t)}{2} dx_t + \frac{1}{2h} \int_{R_t d_t - R_t^A}^{R_t d_t / \lambda - R_t^A} \frac{(R_t^A + x_t) + R_t d_t}{2} dx_t + \frac{1}{2h} \int_{R_t d_t / \lambda - R_t^A}^h \frac{(1 + \lambda)(R_t^A + x_t)}{2} dx_t$$

The first derivative is $(R_t^2/d_t)4h[(\lambda - 1)^2/\lambda - c(\lambda + 1)]$ and the second is $(R_t^2/4h)[(1 - \lambda)^2/\lambda - c(\lambda + 1)]$; both are positive if and only if $(1 - \lambda)^2/\lambda - c(\lambda + 1) > 0$. The condition is satisfied if c is zero, or else if λ and c are sufficiently low. For example, $\lambda < 0.5$ and $c < 0.3$ are jointly sufficient. For our parametrization, this condition is comfortably satisfied.

- Interval E: $R_t^A + h < R_t d_t$. The function reduces to $(1/2h) \int_{-h}^h ((1 + \lambda)(1 - c)(R_t^A + x_t)/2) dx_t$. This is independent of d_t , hence the function is flat and its level is equal to $\frac{1}{2}R_t^A(1 + \lambda)(1 - c)$. Note that the value of the function in this interval is lower than in interval A.
- Interval D: $\lambda(R_t^A + h) < R_t d_t < R_t^A + h$. In this interval the return to outsiders reduces to Eq. (10). Consider this equation in detail. A marginal increase in the deposit ratio has three effects. First, it increases the range of x_t where a run occurs, by raising the upper limit of the first integral; this effect increases the overall return to outsiders by $(1/2h)[((1 + \lambda)(1 - c)/2)R_t d_t]R_t$. Second, it decreases the range of x_t where a run does not occur, by raising the lower limit of the second integral; the effect of this on the return to outsiders is negative and equal to $-(1/2h)R_t^2 d_t$. Third, it increases the return to outsiders for each value of x_t where a run does not occur; this effect is $(1/2h)$

$\left(\int_{R_t d_t - R_t^A}^h \frac{1}{2} dx_t \right) R_t = (1/2h) \left((h - R_t d_t + R_t^A)/2 \right) R_t$. Equating to zero the sum of the three effects and solving for d_t yields Eq. (11). Since the second derivative is negative, this is a local maximum. Note that this local maximum is within interval D if $\lambda < 1/(2 - \lambda + c(1 + \lambda)) < 1$, a condition comfortably satisfied in our case. Given the shape of the function in the other intervals, this is also a global maximum.

The graph below plots the function 9 against d_t , for the following parameter values: $R_t^A = 1.03$; $R_t = 1.005$; $\lambda = 0.45$; $h = 0.45$; $c = 0.2$. For $h < 0.39$, interval C vanishes, unless λ declines sufficiently, but all other properties carry through and the global maximum remains in interval D, as described:



In this interval, the expected payoff to the capitalist, that enters in the bank capital accumulation equation, is equal to

$$R_t^{BK} = \frac{1}{2h} \int_{R_t d_t - R_t^A}^h \frac{(R_t^A + x_t) - R_t d_t}{2} dx_t = \frac{(R_t^A + h - R_t d_t)^2}{8h}$$

Appendix C. Elasticity of the return on assets to changes in the policy rate

To illustrate the mechanics of the risk taking channel in the model, we examine the effect of a contractionary monetary shock on the two key interest rates, R_t and R_t^A ; see chart below (Fig. C1). As we can see that the response of both rates is positive, and in fact the first is larger than that of the second (this is true for plausible parameter values). As we have seen in the main text, this is sufficient to show that bank risk declines.

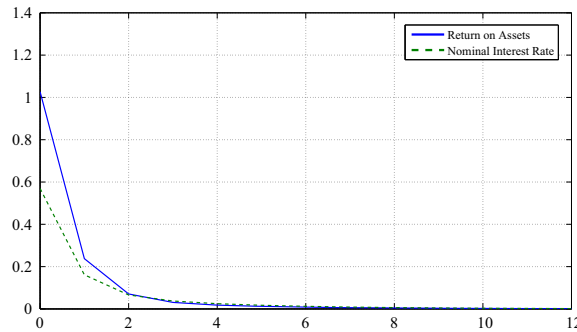


Fig. C1. Impulse responses: return on assets and interest rates to monetary policy shocks.

Appendix D. Impulse response to technology shocks. robustness to different policy rules

In this section we perform robustness checks to verify that the transmission mechanism of the technology remains qualitatively the same across different monetary policy rules. The figure shows impulse response functions to technology shocks under three different monetary policy rules: (a) a standard Taylor rule calibrated as in the benchmark parametrization, (b) a rule with $\phi_\pi = 2, \phi_y = 0.5/4, \phi_r = 0$, (c) a rule with $\phi_\pi = 2, \phi_y = 0.5/4, \phi_r = 0.8$. The transmission mechanism of the technology shock remains the same under the three cases considered with the only exception that under a positive parameter on interest rate smoothing there is a more pronounced hump shaped dynamics (Fig. D1).

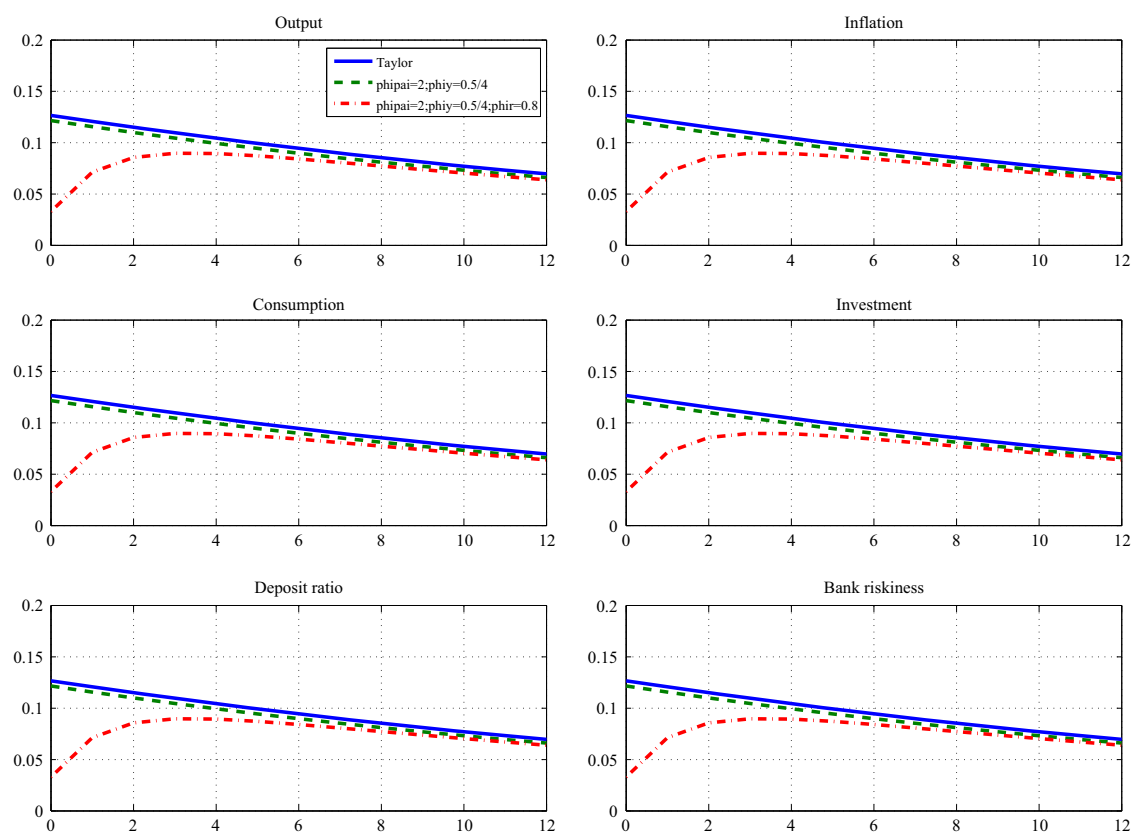


Fig. D1. Impulse responses to positive technology shocks under different monetary policy rules.

Appendix E. Data description

Fig. E1 shows the data used in the VAR estimation.

Variable name	Description
Industrial production	Logarithm of the total industrial production index.
Consumer prices	Logarithm of Consumer Price Index
Commodity price	All items all urban areas. <i>Source</i> : Datastream.
inflation	Logarithm of the Commodity Research Bureau Spot Index. <i>Source</i> : Datastream.
Bank asset risk 1	Logarithm of debt of non-financial corporations. <i>Source</i> : Haver.
Bank asset risk 2	Logarithm of debt of households. <i>Source</i> : Haver.
Monetary policy	Effective Federal Fund rate. <i>Source</i> : Datastream.
Bank funding risk	Sum of REPOs, large time deposits and assets of money market mutual funds. Data for REPOs and large time deposits are taken from the flow of funds statistics for liabilities of depository institutions (Line 110). Data for money market funds assets are from Line 206. <i>Source</i> : Haver.
Bank overall risk	Realized volatility of the returns of the US bank equity Datastream Index. The realized volatility is calculated as the average daily absolute return of the index over each quarter. <i>Source</i> : Authors' calculation and Datastream.

Notes: The order of the variables in the table reflects the order of the variables in the VAR, i.e. the shock to the macro-variable is exogenous, while the shock on bank risk, the last shock, is a combination of all the other shocks. The estimation period of the baseline model is 1980–2008, quarterly data.

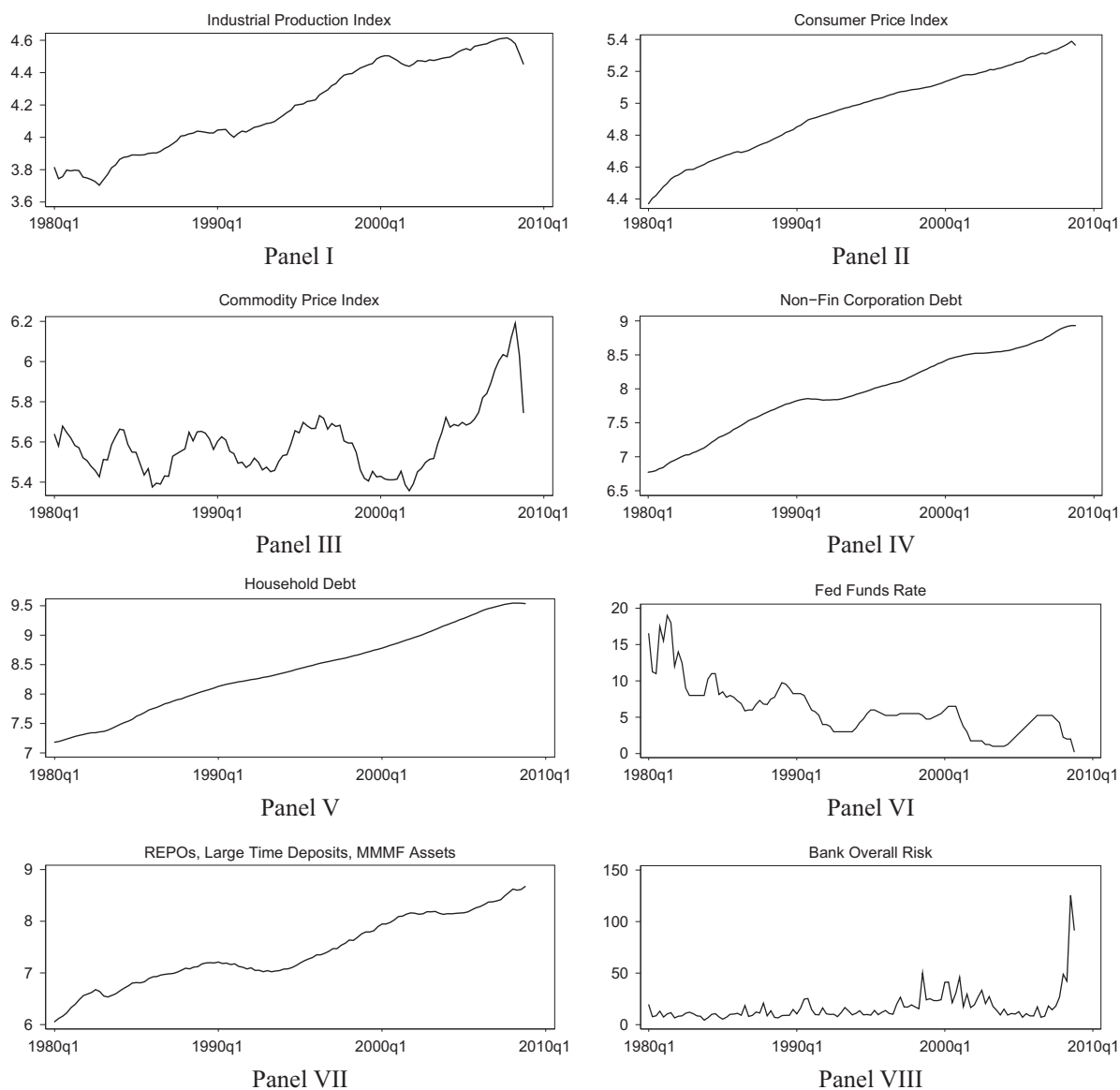


Fig. E1. Data used in the VAR estimation: industrial output (I), consumer prices (II), commodity prices (III), corporate debt (IV), household debt (V), fed funds rate (VI), non-deposit bank liabilities (VII), bank overall risk (VIII).

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